SE Lab – 01

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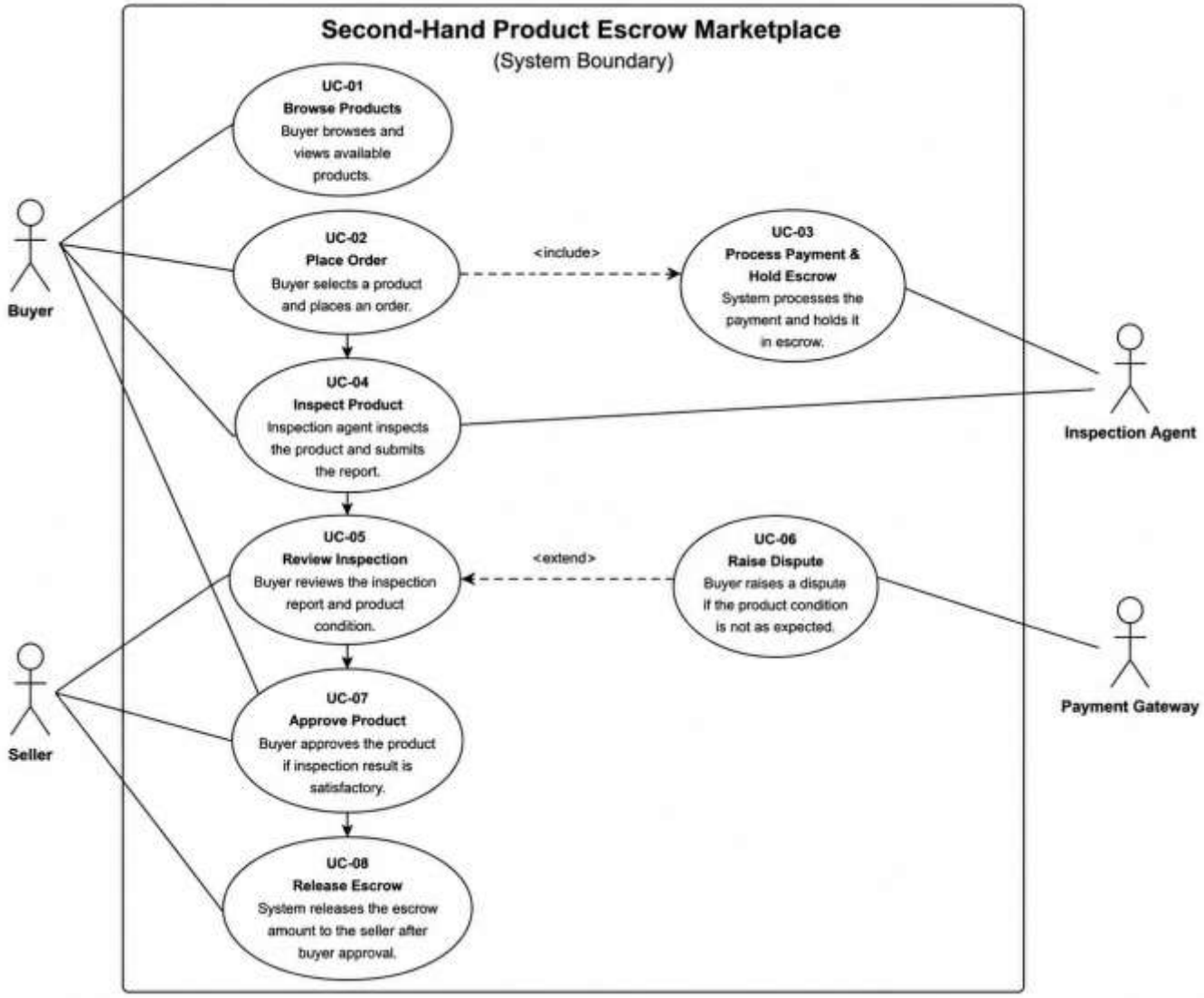
SECOND HAND PRODUCT ESCROW MARKETPLACE

Problem Statement – 39

Derivable 1 – Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria — Pass / Fail	Rationale
FR-001	Functional	The system shall lock the buyer's payment in escrow after successful payment and change the order status to **"Under Inspection"** until an inspection agent submits physical verification sign-off.	High	**Pass:** Payment is successfully held in escrow and order status changes to *Under Inspection* after payment. **Fail:** Payment is released to the seller before inspection sign-off or the status does not change correctly.	Protects the buyer's money until the product has been independently inspected.
FR-002	Functional	The system shall allow a buyer to place an order for an available second-hand product and initiate payment through the supported payment gateway.	High	**Pass:** Buyer selects an available product, places the order, and is directed to payment. **Fail:** Buyer cannot place an order for an available product or payment cannot be initiated.	Provides the basic purchasing functionality required to start an escrow transaction.
FR-003	Functional	The system shall allow an authorized inspection agent to view an assigned order and submit a physical verification report containing the product's condition and inspection result.	High	**Pass:** Authorized inspector submits all required inspection details and the system records the report successfully. **Fail:** An unauthorized user can submit a report or the system accepts a report with required information missing.	Ensures that product condition is independently verified before the buyer makes the final decision.
FR-004	Functional	The system shall allow the buyer to review the inspection report and either approve the product or raise a dispute if the product does not meet the expected condition.	High	**Pass:** Buyer can view the completed inspection report and select either **Approve** or **Raise Dispute**. **Fail:** Buyer cannot access the report or cannot submit either decision.	Gives the buyer control over the final acceptance of the product.
FR-005	Functional	The system shall release the escrowed payment to the seller only after a valid inspection report has been submitted and the buyer has approved the product.	High	**Pass:** Funds are released only when both inspection completion and buyer approval are recorded. **Fail:** Funds are released when either condition is missing.	Prevents premature payment to the seller and maintains the core escrow protection.
NFR-001	Nonfunctional	The system shall immediately freeze escrow funds and prevent withdrawal of the affected transaction amount when a dispute resolution ticket is active.	High	**Pass:** After a dispute is created, escrow funds remain frozen and cannot be withdrawn until the dispute is resolved. **Fail:** Seller receives or withdraws the disputed funds while the investigation is active.	Protects both parties' funds while a transaction is being investigated.
NFR-002	Nonfunctional	The system shall display the current order and escrow status to an authorized user within **2 seconds** under normal operating conditions.	Medium	**Pass:** At least 95% of status requests return the current status within 2 seconds during testing. **Fail:** More than 5% of status requests take longer than 2 seconds.	Ensures that users can quickly see whether payment, inspection, approval, or dispute processing is in progress.

Derivable 2 – Use Case Diagram



Derivable 3 - Use-Case Flow Document

Use Case ID	UC-02 (includes UC-03)
Use Case Name	Place Order & Process Payment
Primary Actor	Buyer — initiates the order and drives the interaction from start to finish.
Secondary Actors	Payment Gateway — external system that authorizes and processes the transaction; Inspection Agent — receives the confirmed order for physical verification.
Preconditions	1) Buyer is registered and logged in. 2) The selected product exists, is listed as "Available," and has not been ordered by another buyer. 3) Buyer has a valid, supported payment method (card / wallet / net-banking) linked or ready to enter.

Postconditions — Success	Payment is successfully authorized and locked in escrow; funds are not released to the Seller. The order status changes to "Under Inspection", and both Buyer and Seller are notified.
Postconditions — Failure	No funds are captured or held in escrow; the order is not created (or is auto-cancelled); the product remains listed as "Available"; Buyer is notified of the failure reason.

Main Success Scenario: -

1. Buyer browses the marketplace and selects a product to purchase.
2. Buyer clicks "Place Order" on the product page.
3. System checks that the product is still available and not already reserved by another order.
4. System displays an order summary showing price, escrow terms, and estimated inspection timeline.
5. Buyer reviews and confirms the order, proceeding to checkout.
6. System triggers the «include» Process Payment use case and redirects the Buyer to the Payment Gateway.
7. Buyer enters payment details and submits the payment.
8. Payment Gateway validates and authorizes the transaction.
9. Payment Gateway sends a payment-success confirmation back to the System.
10. System locks the authorized amount in escrow (funds are held, not released to the Seller).
11. System updates the order status from "Available" to "Under Inspection."
12. System sends confirmation notifications to both the Buyer and the Seller, and assigns the order to an Inspection Agent.

Alternate / Exception Flows :-

A1 — Product No Longer Available (extends step 3)

Triggered when another buyer reserves or purchases the product first.

a) System detects the product status is no longer "Available."
b) System displays an "Item No Longer Available" message to the Buyer.
c) System returns the Buyer to the product listings; no order or payment is created.

A2 — Payment Declined (extends step 8)

Triggered when the Payment Gateway rejects the transaction (e.g., insufficient funds, card declined, gateway timeout).

a) Payment Gateway returns a decline/failure response to the System.
b) System displays a "Payment Failed" message to the Buyer and keeps the order in a "Payment Pending" state.
c) Buyer is prompted to retry with the same or a different payment method.
d) If payment fails twice, the System automatically cancels the order, releases the product back to "Available," and notifies the Buyer that no funds were held in escrow.

What Happens If the Use Case Fails :-

If the use case fails at any point before step 10 (escrow lock), no money moves: the payment is never captured, the order is not created or is cancelled, and the product reverts to "Available" so other buyers can purchase it. The Buyer is always shown a clear failure message stating the reason (item unavailable or payment declined) and, where applicable, is given the option to retry. Because payment is only committed to escrow after explicit gateway confirmation (step 9) and the order status only changes after that (step 11), a failure can never leave the system in an inconsistent state where funds are held but no valid order exists, or an order exists with no payment behind it.